MACHINE LEARNING — DETAILED MASTER INDEX

Detailed Master Index • B.Tech CSE 4th Year • Placement / Interview / OA Preparation

0. COMMON CSE + INTERVIEW FOUNDATION

0.1 Programming [B][INT][OA] — Variables, data types, operators, input/output, conditionals, loops, functions, scope, recursion, exceptions and debugging.

0.2 Python [B][INT][OA] — Lists, tuples, sets, dictionaries, strings, slicing, comprehensions, lambda, map/filter/reduce, iterators, generators and modules.

0.3 OOP [B][INT] — Class/object, constructor, encapsulation, inheritance, polymorphism, abstraction, composition, overriding, overloading concepts and SOLID basics.

0.4 DSA [B→A][INT][OA] — Arrays, strings, hashing, linked lists, stacks, queues, trees, BST, heap, trie, graphs, recursion, backtracking and dynamic programming.

0.5 Algorithms [B→A][INT][OA] — Sorting, binary search, two pointers, sliding window, prefix sums, greedy, divide-and-conquer, BFS/DFS and DP.

0.6 Complexity [B][INT][OA] — Big-O, Big-Theta, Big-Omega, time/space complexity, recursion complexity and amortized analysis.

0.7 SQL [B→A][INT][OA] — SELECT, WHERE, GROUP BY, HAVING, joins, subqueries, CTEs, CASE, windows, ranking, dates, NULLs and query optimization.

0.8 Git/Linux [B][P] — Git workflow, branches, merge/rebase, GitHub, CLI, permissions, processes, pipes, grep/sed/awk and environment variables.

0.9 Interview method [INT] — Clarify → approach → brute force → optimize → code → test edge cases → complexity → alternatives → trade-offs.

1. ML FOUNDATIONS

1.1 ML terminology [B] — Samples, features, targets, labels, parameters, hyperparameters and objectives.

1.2 Learning paradigms [B][INT] — Supervised, unsupervised, semi-supervised and reinforcement learning.

1.3 End-to-end workflow [B][INT] — Problem framing → data → EDA → preprocessing → baseline → model → validation → tuning → deployment.

1.4 Leakage [INT] — Target leakage, train-test contamination, preprocessing leakage and prevention.

1.5 Baselines [INT] — Dummy models, simple heuristics and why baselines matter.

2. MATHEMATICS + STATISTICS [M][INT]

2.1 Linear algebra — Vectors, matrices, dot products, norms, projections, rank, eigen concepts and SVD.

2.2 Probability — Conditional probability, Bayes theorem, independence, expectation and variance.

2.3 Statistics — Descriptive statistics, sampling, distributions, confidence intervals and hypothesis testing.

2.4 Calculus — Derivatives, partial derivatives, gradients and chain rule.

2.5 Optimization — Loss surfaces, gradient descent, learning rate, convexity intuition and convergence.

3. DATA PREPROCESSING + FEATURE ENGINEERING [B→A][INT][OA]

3.1 Cleaning — Missing values, duplicates, inconsistent types and noisy data.

3.2 Outliers — IQR, z-score, robust methods and domain-aware handling.

3.3 Encoding — Label, ordinal, one-hot, frequency and target encoding.

3.4 Scaling — Standardization, normalization and algorithm sensitivity.

3.5 Features — Transformations, interactions, aggregations, temporal features and domain features.

3.6 Selection — Filter, wrapper and embedded methods.

3.7 Pipelines — Reproducible preprocessing and leakage-safe transformations.

4. SUPERVISED LEARNING [B→A][INT][OA]

4.1 Linear regression — OLS, assumptions, coefficients, residuals and diagnostics.

4.2 Regularization — Ridge, Lasso and Elastic Net; effect on coefficients.

4.3 Logistic regression — Sigmoid, log-odds, decision boundaries and regularization.

4.4 k-NN — Distance metrics, k selection, scaling and curse of dimensionality.

4.5 Naive Bayes — Bayes rule, conditional independence and variants.

4.6 Decision trees — Entropy, Gini, information gain, splits, pruning and depth.

4.7 SVM — Margin, support vectors, kernels and C/gamma concepts.

4.8 Ensembles — Bagging, random forest, boosting, AdaBoost, gradient boosting, XGBoost and LightGBM.

4.9 Model comparison — Accuracy vs interpretability, speed, data requirements and failure modes.

5. UNSUPERVISED + ADVANCED ML [I→A][INT]

5.1 Clustering — K-means, hierarchical clustering, DBSCAN and cluster validation.

5.2 Dimensionality reduction — PCA, explained variance and practical use.

5.3 Anomaly detection — Statistical, distance-based and isolation-based approaches.

5.4 Imbalanced learning — Class weights, resampling, SMOTE concepts and proper metrics.

5.5 Time-series ML — Lag features, rolling statistics, temporal splits and forecasting concepts.

5.6 Recommenders — Content-based, collaborative filtering and ranking concepts.

5.7 Explainability — Feature importance, permutation importance, SHAP/LIME concepts.

5.8 Robustness — Distribution shift, concept drift, uncertainty and model stability.

6. EVALUATION + HYPERPARAMETER OPTIMIZATION [INT][OA]

6.1 Regression metrics — MAE, MSE, RMSE, R^2 and MAPE limitations.

6.2 Classification metrics — Accuracy, precision, recall, F1, ROC-AUC and PR-AUC.

6.3 Thresholding — Probability thresholds, cost-sensitive decisions and precision-recall trade-off.

6.4 Cross-validation — K-fold, stratified, grouped and time-series validation.

6.5 Tuning — Grid search, random search and Bayesian optimization concepts.

6.6 Calibration — Reliability of predicted probabilities and calibration methods.

6.7 Error analysis — Slice analysis, confusion patterns, residual analysis and failure taxonomy.

7. ML ENGINEERING + MLOPS [A][P][INT]

7.1 scikit-learn engineering — Estimator API, pipelines, preprocessing, persistence and reproducibility.

7.2 Experiment tracking — Parameters, metrics, artifacts, seeds and comparisons.

7.3 Model serving — Batch vs online inference, REST APIs and serialization.

7.4 Docker — Images, containers, dependencies and reproducible runtime.

7.5 Monitoring — Data drift, concept drift, performance, latency and system metrics.

7.6 Lifecycle — Versioning, model registry, approval, rollback and retraining.

7.7 CI/CD — Testing, packaging, validation and automated deployment.

8. PROJECTS + INTERVIEW/OA [P][INT][OA]

8.1 Project A — Classical supervised ML end-to-end with strong feature engineering and evaluation.

8.2 Project B — Unsupervised/anomaly/recommendation use case with business interpretation.

8.3 Project C — Production ML API with Docker, monitoring and documentation.

8.4 ML case study — Design a model for a business problem and justify every decision.

8.5 Interview bank — Algorithm assumptions, comparisons, metrics, leakage, regularization, CV and tuning.

8.6 OA bank — Python, DSA, SQL, statistics, ML MCQs and timed coding.

Final preparation sequence: Fundamentals → Core theory → Implementation → Problem solving → Projects → Deployment → Technical interview → OA → Mock interview → Revision.